

Evaporation Lab

Pre-Lab Plan
Unit Operations Lab 2
January 23, 2020

Group 1

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Commented [DL1]: 25.5/35
Wrong footer.

Commented [DL2]: Lab 1.

1. Experimental Objective

Evaporation is an equilibrium-based separation process by which a solution consisting of a non-volatile solute and a volatile solvent mostly water brought to a concentration. The product of evaporation is either a heavy liquid or liquor and a vapor consisting of a portion of the solvent. The vapor is usually condensed and removed. Evaporation is distinct from drying, crystallization and distillation as a process in several important ways. Evaporation produces a liquid product (sometimes a very viscous one) which separates it from drying that in turn produces a solid. It puts no emphasis on forming crystals which clearly separates it from crystallization. Finally, it differs from distillation in that the vapor produced is often single component. However, there is no emphasis on or specific step for separation of the vapor mixture if it is a mixture. The objective of this experiment is to model a small-scale evaporation system where a single-effect evaporator unit will be studied by using steam to drive separation of a solvent (water) from an aqueous sugar solution. Parameters such as solvent volume, economy, capacity will be recorded by considering the effects of pressure and temperature. Furthermore, the effect of evaporation on liquid characteristics such as concentration, foaming, scale and temperature sensitivity will be considered throughout the experiment.

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Commented [DL4]: Include the sentence in the introduction section in the final report.

Commented [AZ5]: This is not needed in the prelab report. Use this in your final lab report.

Commented [DL6]: The objective of this lab is to concentrate a solution which consists of a nonvolatile solute (sugar) and a volatile solvent (water).

Commented [DL7]: Adopt different vacuum settings.

Commented [DL8]: Good.

2. Experimental

2.1 Apparatus

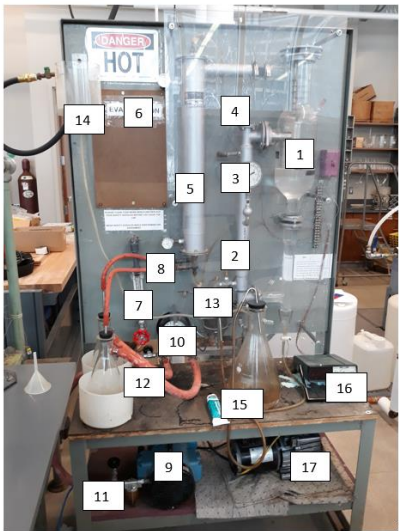


Figure 1: Evaporation Unit

Table 1: Apparatus Equipment

Number	Apparatus	Description
1	Evaporation Vessel	A place a sample is loaded, and evaporation take place
2	Heat Exchanger	A shell and tube heat exchanger
3	Pressure	To measure the pressure change
4	Valve (1)	Used for closing and opening the valve
5	Condenser	Used to condes condense the vapor coming out evaporation vessel
6	Temperature Gauge	To measure the pressure difference
7	Rotameter	Measure the fluid flow through the apparatus
8	Valve(2)	Opening that allow air flow
9	Vacuum Pump	It creates vacuum pressure
10	Vacuum Pressure Gauge	Measure the magnitude of vacuum pressure
11	Valve(3)	Responsible for water flow into the heat exchanger
12	Valve (4)	Drainage of the evaporation vessel
13	Steam Trap	Trap and condes steam from heat exchanger
14	Water Tank	Provide extra water supply
15	Thermocouple	Measure the temperature at specific location

Commented [DL9]: 3/5
A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

Commented [AZ10]: Don't start a section with a figure. You need to provide some text here, and refer to the figures and tables in the text.

Commented [DL11]: Pressure gauge.

Commented [AZ12]: Pressure of what fluid?

Commented [DL13]: How a valve is used for closing and opening the valve?

Commented [DL14]: Unit.

Commented [DL15]: ?

Commented [AZ16]: A temperature gauge doesn't read pressure, it reads the temperature

Commented [DL17]: Unit and the maximum capacity.

Commented [DL18]: Unit.

Commented [AZ19]: condense

Commented [AZ20]: This is not a thermocouple, it looks like the steam product flask.

16	Thermocouple Monitor Reader	Display the measured temperature
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2.2 Materials and Supplies

Table 2: Materials & Supplies

Materials	Description
Water	Forming of the 10 wt% sugar mixture and to keep a level concentration
Low Pressure Steam	Pressured stream supplied to heat exchanger
Sugar	Used to form the 10 wt% sugar mixture
4000 mL Erlenmeyer flask	Collect steam condensate
2 2000 mL Erlenmeyer flask	Collect distillate condensate
Funnel	Used for the transfer of the solution into the evaporation vessel
1000 mL Graduated cylinder	Used to measure the liquid collected
Dow Corning high vacuum grease	Good for extra vacuum sealing on the Erlenmeyer flasks
OMEGA 2166A Digital Thermometer	Measures the temperatures of the 8 different thermal couples
Timer	Measure the amount of time the condensate accumulates
Mass Balance Scale	Measures the weight of the liquid
Scoopula	Used to transfer solids

Commented [DL21]: 4.5/5
Cooling water and heat-resistance gloves are needed.

Commented [AZ22]: You have this in the apparatus section

2.3 Experimental Plan

- Before starting the experiment, be sure to drain the contents from the separator through the valve which is located at the bottom of the separator.
- Prepare a 10 wt% sugar and fill the separator to the level line on the apparatus itself.
- When pouring the sugar solution into the separator, allow the solution to flow inside and used to drain the separator.
- Place pure water in the makeup tank (NOTE: In the experiment, the flow of the makeup water may be adjusted so that the system stays at a steady level.)
- Start the flow of steam.
- Start the cooling water to the steam tap (NOTE: Any remaining from the steam will condense in a graduated cylinder.)
- Turn on the vacuum pump and adjust the bypass valve to get a proper pressure in the separator (NOTE: The pressure gauge reads in inches of mercury.)
- Because the solution will be circulating through the heat exchanger a temperature difference will exist. This experiment will be performed at several vacuum pressure settings due to this phenomenon.
- Collect the vapor product once it has condensed (NOTE: Be vigilant to add enough makeup water to maintain the level at a steady state.)
- Be patient as the system reaches steady state as indicated by the temperature readings.

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Commented [DL24]: How much?

Commented [DL25]: Which one should be the first?

Commented [DL26]: How much?

Commented [DL27]: Have you noticed that the left side and the right side of the gauge have different units?

Commented [DL28]: How much?

Commented [DL29]: What is the sign of the steady state?

- Once the system reaches steady state, indicate the level of the steam condensate and collected vapor.
- Wait roughly 15 – 20 minutes and read the levels again (**NOTE:** Reading these levels again will enable the calculations of the flowrates of the two streams; **measure the time needed to collect a given volume of a fluid.**)
- Document the flow rate of the cooling water to the steam trap, and both the inlet and outlet temperatures (**NOTE:** This will be needed to determine how much of the steam condensed inside the heat exchanger.)
- Change the pump to a different vacuum level for a total of **three vacuum settings** (**NOTE:** The boiling point of the solution will change and should raise the temperature difference between the solution and the steam.)

Commented [DL30]: Be specific.

Commented [DL31]: Be specific.

Project Safety EvaluationProject Title: Evaporation LabDate: 01/23/2020

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: the power supply connected to the vacuum <u>pump</u>
Potential mechanical hazards?	Yes	If yes, identify conditions: Evaporator contains hot fluid, pressurized <u>stream</u>
Potential pressure hazards?	Yes	If yes, identify conditions: High pressure steam coming from heat exchanger unit.
Potential temperature hazards?	Yes	If yes, identify conditions: Hot heat <u>Exchanger</u>
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Nitrile, Heat resistance gloves
MSDS reviewed by all team members?	<u>Yes</u>	List MSDS sheets reviewed:
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	120	In the heat exchanger, It will be hot so avoid <u>contact</u>
Pressure (psia)	<u>40</u>	Heat exchanger and condenser
Safety Controls	yes/no	If yes, Provide description
	Steam shut off valve	It stops and allow the steam flow to heat <u>exchanger</u>
	Main power switch	Can be used to cut power from all the electrical equipment.

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Commented [DL33]: Thermocouples.

Commented [AZ34]: Explain the hazard. Also water hazard above thermocouple connections

Commented [AZ35]: Steam; why is this a hazard?

Commented [AZ36]: The whole apparatus is hot

Commented [DL37]: List the MSDS reviewed.

Commented [AZ38]: The sample flasks may be hot. Also the vessel itself.

Commented [DL39]: 2-15 psia.

Commented [AZ40]: Also cooling water shut-off valve

Commented [AZ41]: yes

I have read relevant background material for the Unit Operations Laboratory entitled:

“_____ Evaporation _____” and
understand the hazards associated with conducting this experiment. I have planned out my experimental
work in accordance to standards and acceptable safety practices and will conduct all of my experimental
work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other
lab students, and will look out for their safety as well.

Signature of Team Member	Date
Marvel Odeesh	1/23/2020
Gewargis Garmooos	1/23/2020
Mahamat Abdraman	1/23/2020
Sinan Masoud	1/23/2020
Yohannes Teweldeberhan	1/23/2020